

From the continuum to discrete substrates: 't Hooft and other turns

Conversation record, 2026-09-28 (user and Claude, with a from-memory answer by Claude Fable). All references below are quoted from memory and have not been verified; they are leads for the prior-art section of the fifth-postulate paper, and no result here is claimed.

Conversation with the user, 2026-09-28 night (not yet in any repo note; references from memory, unverified).

't Hooft (Fable's memory, high/medium confidence): renormalizability and $\beta < 0$ (Marseille 1972); only first two beta coefficients scheme-independent (limits "2 b0 per log" beyond one loop); renormalons (Erice late 1970s) = the atlas exponential ladder, threshold $2 b0 c = 4$ is the gluon-condensate renormalon; instantons $e^{-8 \pi^2/g^2}$, U(1) problem; 't Hooft loop $B(C)$ (NPB 138, 1978) = our mid-edge centre flip; twisted b.c. and e/m flux sectors (NPB 153, 1979), fractional charge $1/N \bmod 1$ ($1/3$ for SU(3), fits the trisection result); the round-12 non-based-cycle Aharonov-Bohm phase is the small-angle part of his flux-sector data (round-14 prompt now asks Astra to state the covariant reference sector by sector); abelian projection 1981; large N 1974. Cellular-automaton interpretation (1988 on, book 2016): $\hbar$ enters as a conversion constant tied to the time grain δt ; the floor comes from information loss (equivalence classes give H bounded below). This is the exact opposite of the user's thesis on the same RG ground: he supplies $\hbar$ from below; we ask whether consistent refinement forces it.

Candidate next Newton premise (proposed, not yet sent to Astra): take 't Hooft's information-loss/equivalence-class construction as the physical premise and ask whether it bounds the recorded Galileo comparison. Contact point: our thermodynamic-records note gave only $\eta \geq A_0 e^{-W/kBT}$, no floor. Put this to an Astra decision round after round 14.

User's remarks: "as soon as you really understand the RG then you don't understand any" (amused); read Wolfram in the eighties thinking "how will he reach a continuum limit", yet the Newton thesis says the continuum limit is problematic from the start, so "a bit bipolar". Claude's reply: same fault line from two sides; the atlas takes the limit and asks what survives.

Wolfram (Caltech ~1979, QCD phenomenology, Fox-Wolfram moments; cellular automata 1981-82; NKS 2002; hypergraph physics 2020). **Cvitanovic** (Kinoshita's group, QED g-2; birdtracks from colour factors, book 2008, useful notation for our SU(N) Brauer-Klimyk sums; chaos via the Feigenbaum-Cvitanovic RG fixed point, ChaosBook).

Other "Damascus" actors listed: divergence-driven: Heisenberg universal length (1938), Snyder quantized spacetime (1947), Yukawa elementary domains, Dirac's lifelong rejection of renormalization. RG masters going deterministic/discrete: T. D. Lee discrete time (1983), Adler trace dynamics (2004), H. B. Nielsen random dynamics, Feynman checkerboard and "Simulating physics with computers" (1982). Geometry to combinatorics: Regge, Penrose spin networks, Finkelstein space-time code (1969), Sorkin causal sets, Ambjorn-Loll, Wheeler "it from bit", von Weizsacker ur-alternatives. **Closest to the Newton thesis:** Gisin and Del Santo (~2019-21), finite information makes classical mechanics indeterministic (denial of the *velocitas ultima* from quantum information) but give no $\hbar$; the fifth-postulate paper should cite and answer them. Proposed paper sentence: field theorists went discrete to escape infinities; RG masters to explain the quantum; Gisin/Del Santo to break determinism; the user's thesis keeps the continuum and shows consistent refinement demands the constant Newton's limit dropped.

Related: STATE, fifth postulate, stochastic route, thermodynamic records.

't Hooft's automaton against Zeno, the atomists and Newton

Added 2026-09-29 (Claude), at the user's request. The automaton papers are identified by DOI (Crossref-verified, metadata only): the 1988 equivalence relations, the 1999 dissipative deterministic system and the 2016 book. What follows is a reading of his position as Fable and Claude remember it, set against the classics entries of the Planck paper §§7 and 9.

His commitments. Reality is a deterministic automaton with a universal time step δt . A reversible automaton is a permutation of states, so its evolution operator has eigenvalues on the unit circle and energies defined only modulo $2\pi\hbar/\delta t$: $\hbar$ is the conversion between the step and an energy, and the spectrum has no ground state. Information loss, many states merging into one equivalence class, is what he invokes to obtain a Hamiltonian bounded below. Quantum mechanics is the description of the classes; Bell correlations are paid for by superdeterminism.

- **Zeno and Diodorus.** At a tick the automaton holds a configuration and motion exists only as the update between ticks: Diodorus’s “a thing never is moving, but it has moved”, made into dynamics. The Planck paper keeps Zeno’s conclusion for local records only, below the window $\tau_{\text{arrow}} = 8z^2\kappa/(mv^2)$, which depends on the body and the confidence. The automaton makes the arrow exact at every scale below δt and for every body.
- **Democritus and the kalam atom.** The automaton’s cells and ticks are atoms of place and time, the kalam position on time and a stepped cone in Democritus’s dilemma. The Planck paper denies the universal time atom on the evidence of records: its mesh moves with F , m and the confidence, a dynamical resolution. The two are compatible: a universal grain far below every mark mesh leaves Theorem 9 untouched. They differ in status: the grain is ontological, the mesh is epistemic.
- **Epicurus.** Continuity presented to sense, succession below it, and a warning against extrapolating continuity downward: this is ’t Hooft’s picture with the Planck time in place of the threshold of sense. Epicurus is the closest ancient statement of it, as he is of ours; the difference is where the boundary is drawn and whether it is derived.
- **Newton.** The automaton removes the *velocitas ultima* by removing the limit: velocities are finite differences and no instant carries one. That is a third denial of joint determinacy, beside quantum noncommutativity (Theorem A) and stochastic paths (the stochastic route), and it keeps determinism. Its $\hbar$ is a unit conversion, so the automaton alone supplies a grain and no action floor on records: a reversible automaton is a discrete Liouville dynamics, and Theorem I of the unit-and-indeterminacy note gives classical readouts no floor. The floor, in his construction, rides on information loss, and the thermodynamic-records note finds that loss yields a trade-off $\eta \geq A_0 e^{-W/k_B T}$ and no floor.

The sharp question this leaves, the candidate Newton premise above: does coarse-graining by equivalence classes, of the kind that gives his Hamiltonian a ground state, force a positive floor on the recorded inertial–parabola comparison, or only a trade-off? The repository’s current results point to the second; a theorem either way would place the automaton exactly relative to the thesis that quantization is a consistency condition of the continuum limit.

Later the same night: superdeterminism and Dedekind’s cut

Superdeterminism (user: “classical mechanics is accidentally superdeterminist too, and we think it has some mistake somewhere”). From memory: Bell called the independence of settings and hidden variables “free variables” (exchange with Shimony, Horne and Clauser, around 1976–77) and named the escape “super-deterministic” in a 1985 BBC interview (*The Ghost in the Atom*, 1986); Brans gave a model in 1988; ’t Hooft adopted it in the 2000s; Hossenfelder and Palmer revived it in 2020. The link the user and Claude found interesting is developed in the superdeterminism note: for chaotic setting devices, a superdeterministic correlation must be stored in structure of the initial data finer than any fixed resolution, so a floor gives it an Ehrenfest-type lifetime and $\hbar \rightarrow 0$ reopens the loophole.

Dedekind’s cut (user: “Dedekind cuts are peculiar, as they define a point as a segment, really”). Dedekind (1872) defines a real number by the two segments of rationals it separates: the point is given by what lies on either side, and fixing it needs infinitely many comparisons. The intuitionists, and after them Gisin and Del Santo (from memory: finite-information quantities

whose digits are fixed progressively, around 2019–21), refuse the completed cut: a real is only ever a nested sequence of segments, never a point. The repository has a quantitative form of the remark. Locating an instant by nested cuts spends Newton’s cell action additively (cut-measure note, Theorem 2), and with a floor κ per exhibited cut only K_τ/κ cuts can be exhibited (Corollary 5): a recorded instant is a segment of length of order the mark mesh τ_* . Dedekind’s point is the completed limit of that sequence, which is Newton’s side in the scholium closing Book I (Euclid X against least magnitudes); the recorded point is the unfinished one.

Consequence for STATE

None yet. The candidate Newton premise (information loss as in ’t Hooft’s equivalence classes) goes to an Astra decision round after round 14.